



AQ8.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Consider two vectors $a = (2, 0, 3, 0, 8)$, $b = (3, 2, -2, 4, 0)$ in R^5 .

Choose the set of correct options.

☐

a and b are orthogonal.

☐

a and b are not orthogonal.

☐

$(a - b) \cdot a = 0$.

☐

$(a - b) \cdot a = 77$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

a and b are orthogonal.

$(a - b) \cdot a = 77$.

1 point

Choose the set of correct statements.

☐

In an orthogonal set, the norms of all the vectors are equal.

☐

In an orthogonal set, the vectors are linearly independent.

☐

In an orthogonal set, the vectors are linearly dependent.

☐

If the columns of an $n \times n$ coefficient matrix A comprises of the individual vectors of an orthogonal set in R^n , then there must be a unique solution to the system $AX = b$, where X, b are $n \times 1$ vectors.

☐

If the columns of an $n \times n$ coefficient matrix A comprises of the individual vectors of an orthogonal set in R^n , then there are no solutions to the system $AX = b$, where X, b are $n \times 1$ vectors.

☐

The determinant of a square matrix formed by a set of orthogonal vectors in R^n is zero.

☐

A set of n vectors can never form an orthogonal basis in R^{n-1} .

No, the answer is incorrect.

Score: 0

Accepted Answers:

In an orthogonal set, the vectors are linearly independent.

If the columns of an $n \times n$ coefficient matrix A comprises of the individual vectors of an orthogonal set in R^n , then there must be a unique solution to the system $AX = b$, where X, b are $n \times 1$ vectors.

A set of n vectors can never form an orthogonal basis in R^{n-1} .

1 point

Which of the following is an orthogonal basis of the given vector spaces with respect to the standard inner product (dot product)?

☐


$\{(1, 0), (0, 1)\}$ $\{(1, 0), (0, 1)\}$ is an orthogonal basis of R^2 .

☐

$\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is an orthogonal basis of R^3 .

☐

$\{(3, 4), (4, -3), (2, -3)\}$ $\{(3, 4), (4, -3), (2, -3)\}$ is an orthogonal basis of R^2 .

☐

$\{(2, 1, -1), (-1, 1, -1), (3, -3, 3)\}$ $\{(2, 1, -1), (-1, 1, -1), (3, -3, 3)\}$ is an orthogonal basis of R^3 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 0), (0, 1)\}$ $\{(1, 0), (0, 1)\}$ is an orthogonal basis of R^2 .

$\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is an orthogonal basis of R^3 .

Level 2

1 point

Find a vector in R^4 that is orthogonal to the subspace spanned by $(1, 1, 0, 0)$ and $(0, 1, 1, 0)$ with respect to the dot product as the inner product.

☐ $(1, -1, 1, 0)$
☐ $(2, 3, 4, 5)$
☐ $(1, -1, 1, 1)$
☐ $(1, 1, 1, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, -1, 1, 0)$

$(1, -1, 1, 1)$

1 point

Which of the following are orthogonal to the vector $(1, 2)$ in R^2 with respect to the inner product $\langle v, w \rangle = v_1 w_1 - v_1 w_2 - v_2 w_1 + 4v_2 w_2$?

☐

$(-7, 1)$

☐

$(6, -1)$

☐

$(7, 1)$

☐

$(-9, 1)$

☐

$(14, 2)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(7, 1)$

$(14, 2)$

Consider the system of linear equations:

$$x_1 - 2x_2 + 3x_3 = 1$$

$$2x_1 + x_2 = 5 \quad x_1 + x_2 = 5$$

$$3x_1 - 6x_2 - 5x_3 = 9 \quad 3x_1 - 6x_2 - 5x_3 = 9.$$

Number of solutions of the above system is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

1 point

Let AA be the coefficient matrix of the system of linear equations:

$$x_1 - 2x_2 + 3x_3 = 1 \quad x_1 - 2x_2 + 3x_3 = 1$$

$$2x_1 + x_2 = 5 \quad x_1 + x_2 = 5$$

$$3x_1 - 6x_2 - 5x_3 = 9 \quad 3x_1 - 6x_2 - 5x_3 = 9.$$

Which one of the following is true about the matrix AA^T AA^T ?

- ☐ A scalar matrix.
- ☐ The identity matrix.
- ☐ A diagonal matrix.
- ☐ A lower triangular matrix.
- ☐ An upper triangular matrix.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A diagonal matrix.

A lower triangular matrix.

An upper triangular matrix.

[Check Answers](#)

Your score is: 0/7